Day 4 – Array Problems (Reverse Array + Min/Max)

1 Reverse Array

Reverse array ka matlab hai array ke elements ko ulta kar dena.

Example

Input

1 2 3 4 5

Output

5 4 3 2 1

Method 1 – Two Pointer Technique

Steps

Start pointer = first element

End pointer = last element

Dono ko swap karo

Pointers ko move karo

Java Code – Reverse Array

```
public class ReverseArray {

    public static void main(String[] args) {

        int[] arr = {1,2,3,4,5};

        int start = 0;
        int end = arr.length - 1;

        while(start < end){

            int temp = arr[start];
            arr[start] = arr[end];
            arr[end] = temp;

            start++;
            end--;
        }

        for(int i = 0; i < arr.length; i++){
            System.out.print(arr[i] + " ");
        }

    }

}
```

Output

5 4 3 2 1

Time Complexity

O(n)

Space Complexity

O(1)

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2 Find Maximum Element

Maximum element ka matlab hai array ka sabse bada element.

Example

Input
5 9 2 7

Output

9

Java Code – Find Maximum

```
public class MaxElement {

    public static void main(String[] args) {

        int[] arr = {5,9,2,7};

        int max = arr[0];

        for(int i = 1; i < arr.length; i++){

            if(arr[i] > max){
                max = arr[i];
            }

        }

        System.out.println("Maximum element = " + max);

    }

}
```

Output

Maximum element = 9

3 Find Minimum Element

Minimum element ka matlab hai array ka sabse chhota element.

Example

Input
5 9 2 7

Output

2

Java Code – Find Minimum

```
public class MinElement {

    public static void main(String[] args) {

        int[] arr = {5,9,2,7};

        int min = arr[0];

        for(int i = 1; i < arr.length; i++){

            if(arr[i] < min){
                min = arr[i];
            }

        }

        System.out.println("Minimum element = " + min);

    }

}
```

Output

Minimum element = 2

4 Combined Program (Min + Max)

```
public class MinMax {

    public static void main(String[] args) {

        int[] arr = {5,9,2,7};

        int min = arr[0];
        int max = arr[0];

        for(int i = 1; i < arr.length; i++){

            if(arr[i] > max){
                max = arr[i];
            }

            if(arr[i] < min){
                min = arr[i];
            }

        }

        System.out.println("Max = " + max);
        System.out.println("Min = " + min);

    }

}
```

Output

Max = 9
Min = 2

Time Complexity

Operation	Complexity
Reverse Array	O(n)
Find Max	O(n)
Find Min	O(n)

5 Day 4 Practice Problems

- ✓ **Problem 1** Array reverse karo.
- ✓ **Problem 2** Maximum element find karo.
- ✓ **Problem 3** Minimum element find karo.
- ✓ **Problem 4** Array ka first aur last element swap karo.